

Literature Status: Tensor-GUE Strong Convergence and the Peterson–Thom Conjecture

Run fa_banach_001 — lane 14

9 August 2026

Status: literature already answered. Hayes reduces the Peterson–Thom conjecture to strong convergence of two same-size independent GUE families acting on the two tensor factors. Belinschi and Capitaine prove exactly this strong convergence in arXiv:2205.07695, and explicitly note the resulting solution of Peterson–Thom.

1 Original conjecture and reduction

B. Hayes, *A random matrix approach to the Peterson–Thom conjecture*, arXiv:2008.12287 [1], defines independent $k \times k$ normalized GUE families $X^{(k)} = (X_i^{(k)})_{i=1}^r$ and $Y^{(k)} = (Y_i^{(k)})_{i=1}^r$. Theorem 1.1(iii), on arXiv PDF page 3, isolates the following statement; Section 4 restates it as a conjecture.

Source random-matrix conjecture. *For every noncommutative polynomial P ,*

$$\begin{aligned} & \|P(X^{(k)} \otimes I_k, I_k \otimes Y^{(k)})\| \\ & \longrightarrow \|P(s \otimes 1, 1 \otimes s)\|_{C^*(s) \otimes_{\min} C^*(s)} \end{aligned}$$

in probability, where s is a free semicircular family.

Theorem 1.1 proves that this statement implies positivity of the relevant 1-bounded entropy for every nonamenable subalgebra, which in turn implies the Peterson–Thom conjecture: every diffuse amenable subalgebra of a free group factor lies in a unique maximal amenable subalgebra.

2 Separate later answer

S. T. Belinschi and M. Capitaine, *Strong convergence of tensor products of independent G.U.E. matrices*, arXiv:2205.07695 [2], study independent normalized GUE families X_N and Y_N of the same size. Their Theorem 1 on arXiv PDF page 3 states:

Later theorem. *Almost surely, for every noncommutative polynomial P ,*

$$\begin{aligned} & \|P(X_N \otimes I_N, I_N \otimes Y_N)\| \\ & \longrightarrow \|P(s \otimes 1, 1 \otimes t)\|_{\min}, \end{aligned}$$

where s and t are free semicircular systems in the two tensor factors.

This is the source conjecture, with the harmless flexibility of allowing the two family lengths to differ, and with almost-sure convergence rather than merely convergence in probability. The answering paper’s abstract and the discussion following Theorem 1 explicitly state that Hayes’s theorem makes this a proof of the Peterson–Thom conjecture. It also notes an independent Haar-unitary proof by Bordenave and Collins.

Search scope

A bounded check through 9 August 2026 used the run registry, exact title and conjecture wording, tensor-GUE strong convergence, and Peterson–Thom status. The decisive direct match is arXiv:2205.07695. Later work, including arXiv:2308.14109, explicitly calls this the random-matrix solution. No new mathematical result is claimed in this packet.

References

- [1] B. Hayes, *A random matrix approach to the Peterson–Thom conjecture*, Indiana Univ. Math. J. **71** (2022), 1243–1297; arXiv:2008.12287.
- [2] S. T. Belinschi and M. Capitaine, *Strong convergence of tensor products of independent $G.U.E.$ matrices*, arXiv:2205.07695, version 2, 2024.